Assignment 1 - Git & GitHub Fundamentals

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Repository: git-session1-marwanehmamouchi05 on GitHub

1. Repository Setup

The provided starter files were extracted into git-session1-marwanehmamouchi05. The command **git init -b main** initialized the repository. The ignore rules exclude local.env and ordinary .log files, then exempt important.log. The important log is tracked; local.env, app.log, and debug.log are untracked and ignored.

```
local.env
*.log
!important.log
```

2. README & Commit History

The README includes my name, GitHub username, project description, HTML/CSS/JavaScript technologies, a clone command block, an official Git documentation link, and the provided SVG banner. Commits separate the starter, ignore rules, README, status message, CSS styling, About feature, merge, GitHub edit, and final report/evidence.

3. Selective Staging

app.js and style.css were modified together. Only **git add app.js** was run first. The staged diff showed app.js, while the unstaged diff showed style.css. Commit 8e301ce recorded only app.js; style.css remained modified and was committed separately as 7470931. Captured outputs appear on page 3.

4. Branch & Merge

git switch -c feature/about created the About branch. An About section was added to index.html and committed as ad826cb. After **git switch main**, **git merge --no-ff feature/about** created merge commit 947adfd. Both main and feature/about were pushed to GitHub; the history retains the feature branch.

Remote changes, AI use & reflection

5. Remote Changes

An empty public GitHub repository was created, connected as origin, and main was published using **git push -u origin main**. A commit records a snapshot locally; a push sends commits to a remote repository. README.md was then edited directly in the GitHub web editor and committed as f0f362d.

git fetch origin downloaded the remote commit and updated origin/main without integrating it into main. The status showed **behind 1**; **git rev-list --left-right --count main...origin/main** returned **0 1**. SHA-256 hashes of all project files and local HEAD were unchanged. **git pull --ff-only origin main** then fetched and integrated the remote change through a fast-forward.

6. Fair Use of AI

I used Codex to help complete this lab, including executing the Git workflow, editing the starter project, creating the GitHub repository, and drafting this report. My prompt asked it to complete the assignment from the PDF and starter ZIP; I then supplied my name, student ID, and course details, and asked to receive the repository link so I could submit it myself. The generated project changes, commands, and report draft were retained. Codex verified its output against current official Git documentation, actual status/diff/history output, unchanged file hashes after fetch, and the successful pull and push. These checks were performed by the assistant; I still need to review the submission and understand each command before submitting it.

7. Sources & Reflection

Sources: the supplied Assignment 1 instructions/report template and starter ZIP; official Git documentation, accessed through Context7 on October 14, 2026:

Git tutorial (staging and commits): https://git-scm.com/docs/gittutorial

Git diff: https://git-scm.com/docs/git-diff

Git ignore rules: https://git-scm.com/docs/gitignore

Git fetch: https://git-scm.com/docs/git-fetch

Git pull: https://git-scm.com/docs/git-pull

The main concept to focus on is the distinction between the working tree, staging area, local branch, and remote-tracking branch, because each command acts on a different state. If Git behaves unexpectedly, the first checks should be git status, the current branch and upstream, and the appropriate staged or unstaged diff.

Troubleshooting

A deliberate attempt to stage app.log produced Git's ignored-file warning. git check-ignore -v app.log identified the *.log rule. The resolution was to leave this ordinary log untracked, rather than force-add it; important.log remained tracked through the explicit exception.

Evidence: staging, ignore rules & fetch

Selective staging: one staged, one unstaged

```
M  app.js
 M style.css
STAGED DIFF
diff --git a/app.js b/app.js
index 603d28a..e6afb48 100644
--- a/app.js
+++ b/app.js
@@ -1,5 +1,5 @@
-const statusMessage = document.getElementById("status-message");
-
-if (statusMessage) {
-  statusMessage.textContent = "Starter project is ready.";
-}
+const statusMessage = document.getElementById("status-message");
+
+if (statusMessage) {
+  statusMessage.textContent = "Session 1 project is ready: Git workflow completed.";
+}
UNSTAGED DIFF
diff --git a/style.css b/style.css
index b93e076..e9c8817 100644
--- a/style.css
+++ b/style.css
@@ -19,3 +19,8 @@ section {
   padding: 20px;
   border: 1px solid #e3e3e3;
 }
+
+#status-message {
+  color: #216e39;
+  font-weight: bold;
+}
```

After committing app.js only

```
 M style.css
8e301ce Update project readiness message independently of styling
 app.js | 10 +++++-----
 1 file changed, 5 insertions(+), 5 deletions(-)
```

Ignore verification

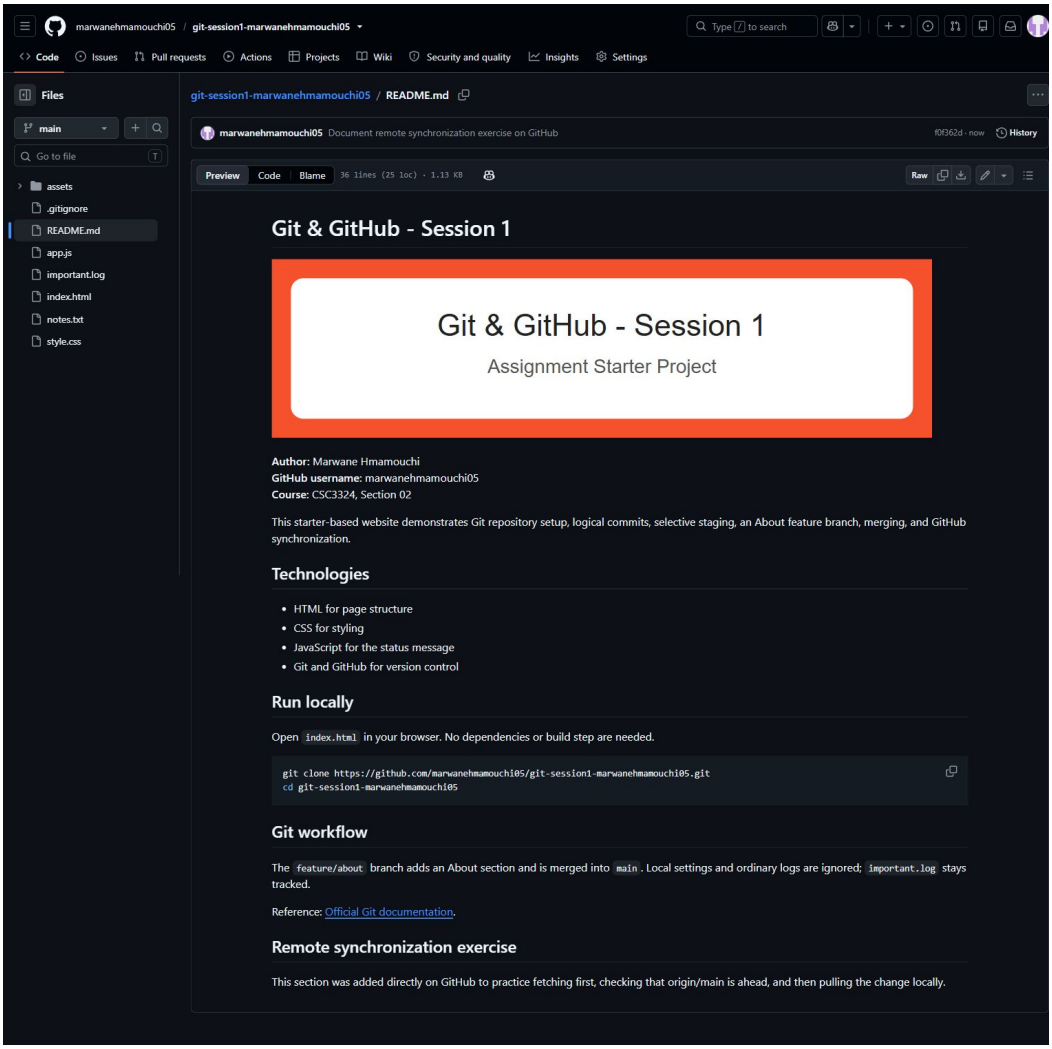
```
!! app.log
!! debug.log
!! local.env
.gitignore:1:local.env■local.env
.gitignore:2:*.log■app.log
.gitignore:2:*.log■debug.log
.gitignore:3:!important.log■important.log
important.log
```

After fetch, before pull

```
From https://github.com/marwanehmamouchi05/git-session1-marwanehmamouchi05
   947adfd..f0f362d  main       -> origin/main
## main...origin/main [behind 1]
Ahead / behind counts:
0■1
f0f362d Document remote synchronization exercise on GitHub
File hashes unchanged; local HEAD unchanged.
```

Evidence: GitHub README & branch history

Rendered README after the web edit



Compact history after pull (before report commit)

```
* f0f362d (HEAD -> main, origin/main, origin/HEAD) Document remote synchronization exercise on GitHub
*   947adfd Merge About feature into main
|\
| * ad826cb (origin/feature/about, feature/about) Add About section describing author and lab
objectives
|/
* 7470931 Highlight project readiness status in green
* 8e301ce Update project readiness message independently of styling
* 4d2ab88 Document author, technologies, setup, and Git workflow
* 228e663 Ignore local settings and ordinary logs; keep important log
* 8bf9dce Add provided Session 1 starter project
```

Full captured command outputs are included in evidence/. The final report commit is added after the history shown above, and pushed to main.